IEM KOLKATA

Linear Algebra
Matrices & Determinants

Q1) If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, then $\det(A^{-1})$ is

(a) 0.25 (b) 0.50 (c) 0.75 (d) 1.25

Q2) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$.

Let det(A) and det(B) denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?

(a) det(A) = det(B)
(b) det(B) = - det(A)
(c) det(A) = 0
(d) det(AB) = det(A) + det(B)

Q3) The matrix $A = \begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$ has det(A) = 100 and trace(A) = 14. The value of $\mid a - b \mid$ is

(a) 1 (b) 2 (c) 3 (d) 4

Q4) If any two columns of a determinant $= \begin{vmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{vmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is CORRECT?

(A) Absolute value remains unchanged but sign will change
(B) Both absolute value and sign will change
(C) Absolute value will change but sign will not change
(D) Both absolute value and sign will remain unchanged

Q5) Consider the following two statements with respect to the matrices $A^{m\times n}, B^{n\times m}, C^{n\times n}, D^{n\times n}$.

Statement 1 : $\mathrm{tr}(AB) = \mathrm{tr}(BA)$
Statement 2 : $\mathrm{tr}(CD) = \mathrm{tr}(DC)$
where tr( ) represents the trace of a matrix. Which one of the following holds?
(A) Statement 1 is correct and Statement 2 is wrong
(B) Statement 1 is wrong and Statement 2 is correct
(C) Both Statement 1 and Statement 2 are correct
(D) Both Statement 1 and Statement 2 are wrong

Q6) Let $ABC = I$, then $B^{-1} = ?$

Q7) Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = I$, then $B^{-1} = ?$

Q8) X, Y are singular matrices $\ni XY = Y$ and $YX = X$. Then show that $X^2 + Y^2 = X + Y$.

Q9) If $| A^{3\times3} | = 3$, then $| (2A)^{-1} | = ?$

Q10) If $| A^{3\times3} | = 4$, then $adj(adj(A)) = ?$

Q11) Consider the system of simultaneous equations:

$$x + 2y + z = 6$$

$$2x + y + 2z = 6$$

$$x + y + z = 5$$

The system has:
(a) unique solution
(b) infinite no of solutions
(c) no solution
(d) exactly two solutions

Q12) Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?
(A) There exist at least $m - n$ linearly independent solutions to this system
(B) There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors
(C) There exists a non-zero solution in which at least $m - n$ variables are 0
(D) There exists a solution in which at least n variables are non-zero

Q13) Consider the system of simultaneous equations:

$$\begin{bmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ 5 \\ 7 \end{bmatrix}$$

Notice that the second and the third columns of the coefficient matrix are linearly dependent. For how many values of α, does this system of equations have infinitely many solutions?
(a) 0 (b) 1 (c) 2 (d) infinitely many

Q14) Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3\times3}$ and $Q \in \mathbb{R}^{3\times1}$.
Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Which of the following statement(s) is/are TRUE?

(a) The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.

(b) If P is invertible, then both L and U are invertible.

(c) If P is singular, then at least one of the diagonal elements of U is zero.

(d) If P is symmetric, then both L and U are symmetric.